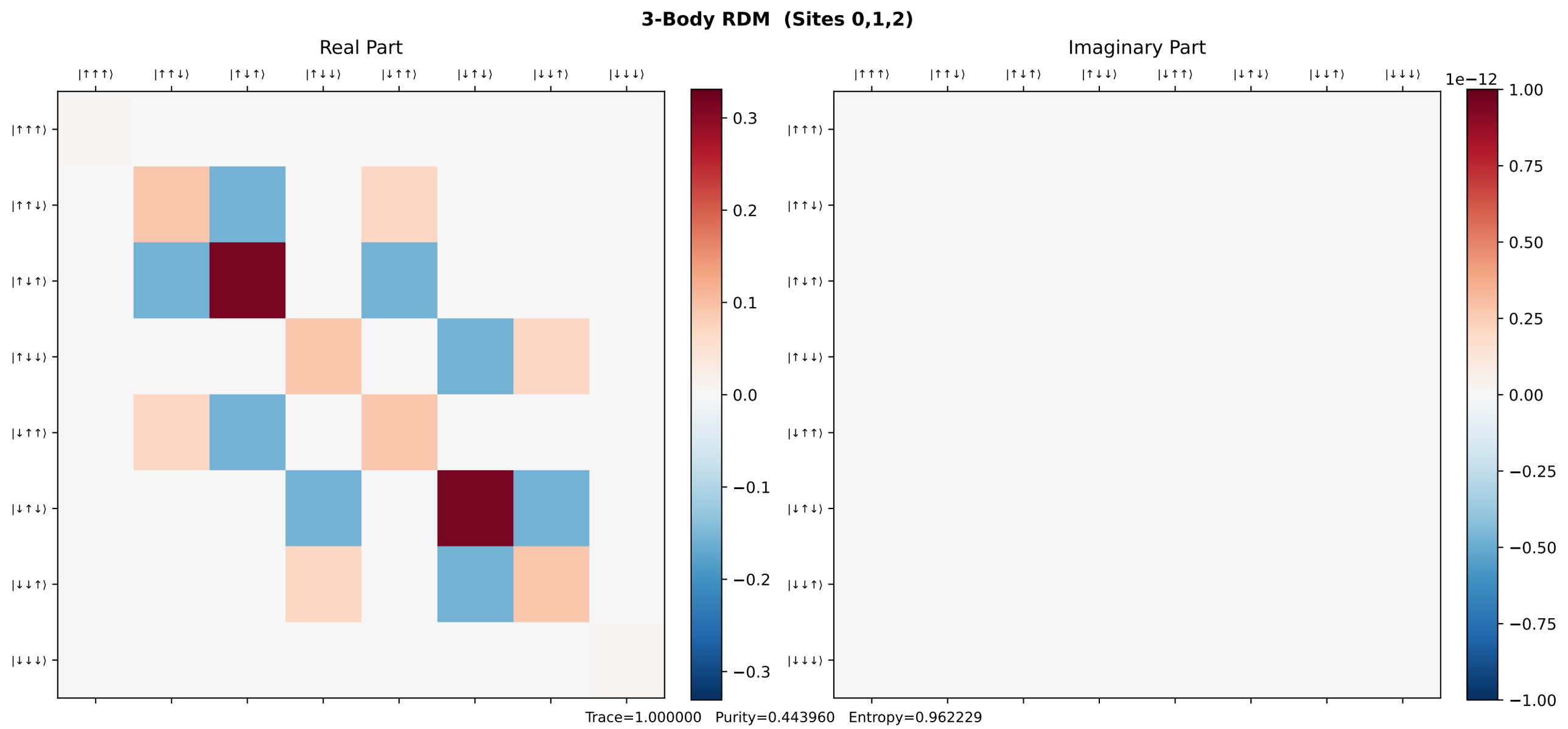
# Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E₀                 : -0.04292319  Extracted n-spin RDM : n = 3
                                   E₁                 : -0.03609210
System parameters                                                    Purity:
-----------------                  ⟨S²⟩               : -0.000000      γ = Tr(ρ²)
Sites (L)          : 6             S                  : -0.0000
Electrons (Ne)     : 6             ⟨Sz⟩               : 0.000000    Separable cut criterion:
Hopping (t)        : 0.1           Multiplicity 2S+1  : 1.0           γ_A = 1  ↔  ρ = ρ_A ⊗ ρ_B
Interaction (U)    : 4.0           Spin state         : Singlet (S=0)
Mode               : heisenberg                                      C_GME (Troiani et al.):
n-body order       : 3             Double occupancy                    C_GME = √(2 · min_A (1 − Tr(ρ_A²)))
                                   ----------------
                                   ⟨D⟩ total          : 0.010679     Decision logic:
                                   ⟨D⟩ / site         : 0.001780       γ = 1  and  all Tr(ρ_A²) < 1   → GME
                                                                       γ = 1  and  some Tr(ρ_A²) = 1  → No GME
                                                                       γ < 1                          → Mixed (convex-roof needed)

                                                                    Date: 2026-03-04  15:07
```


3-Body RDM (Sites 0,1,2)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 0,1,2)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 0,1,2)

## Bipartitions (Sites 0,1,2)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

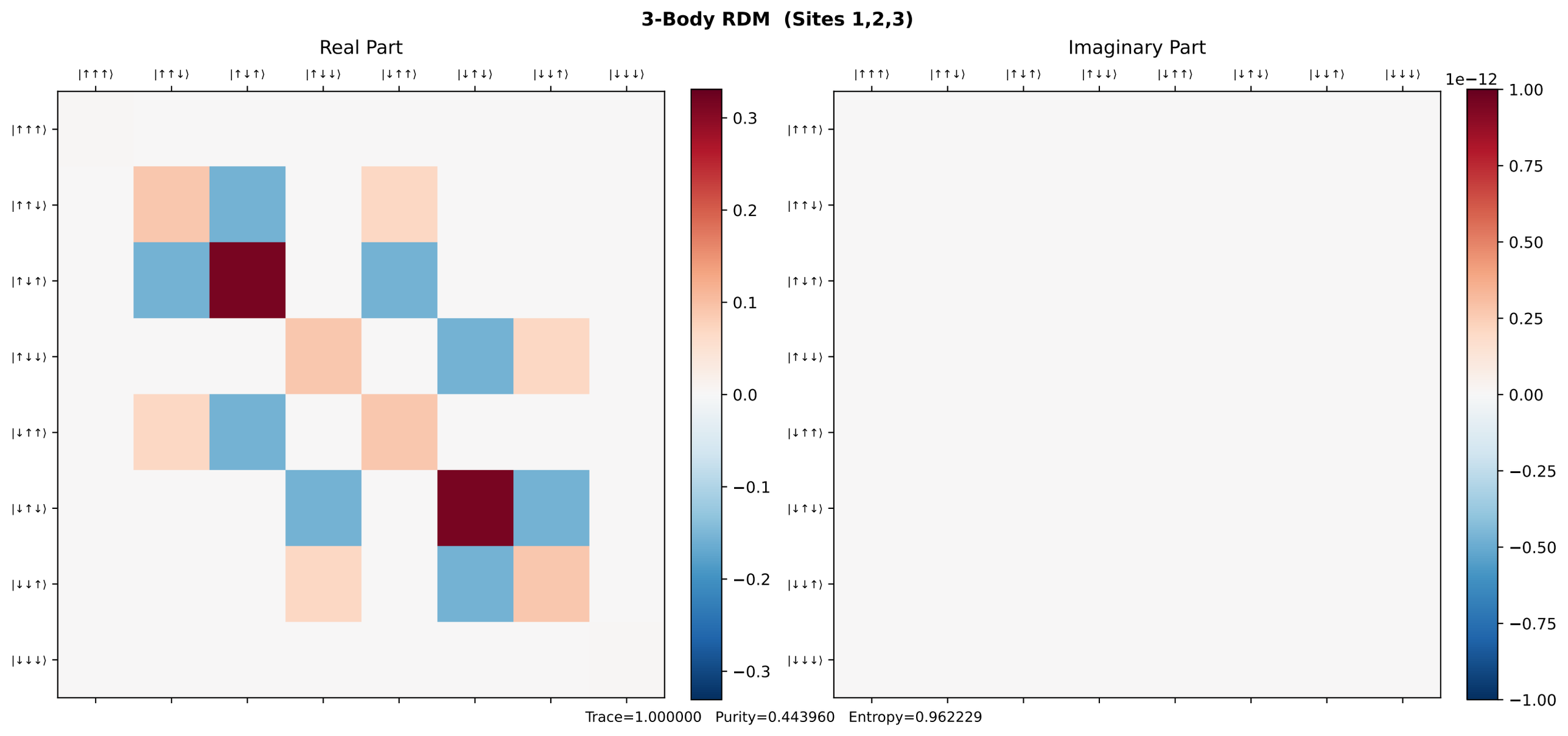
| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 1,2,3)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 1,2,3)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 1,2,3)

## Bipartitions (Sites 1,2,3)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

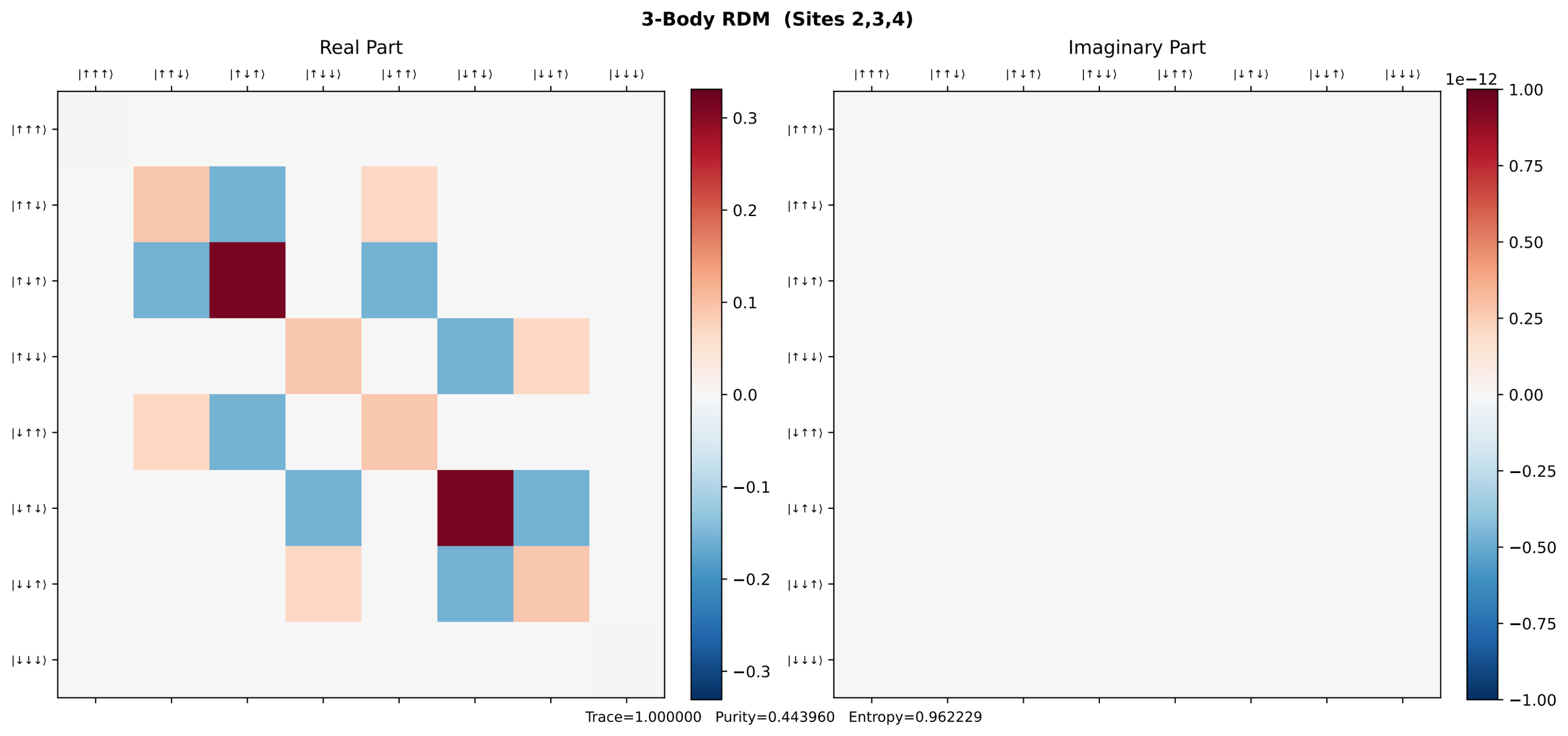
| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 2,3,4)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 2,3,4)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 2,3,4)

## Bipartitions (Sites 2,3,4)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

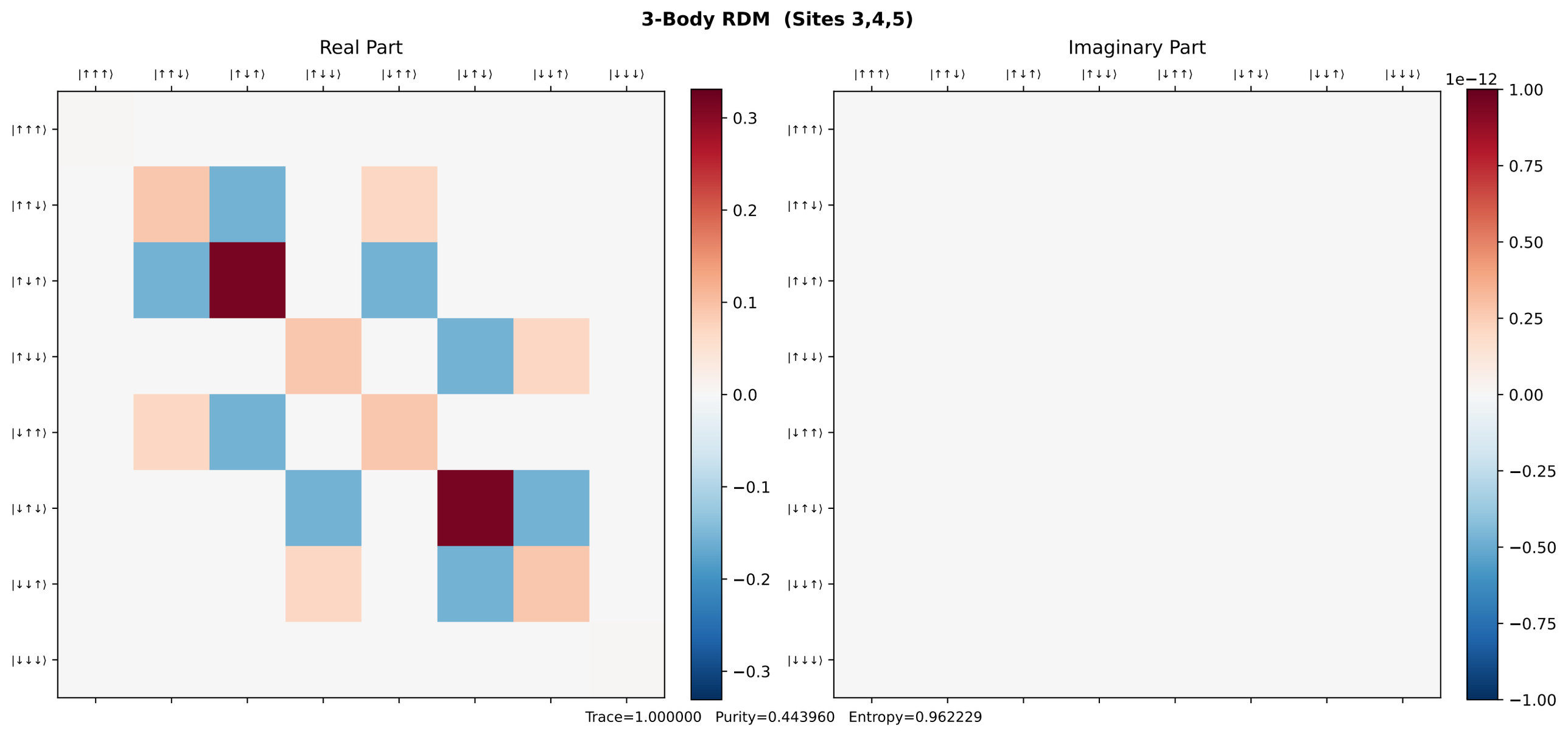
| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 3,4,5)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 3,4,5)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 3,4,5)

## Bipartitions (Sites 3,4,5)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

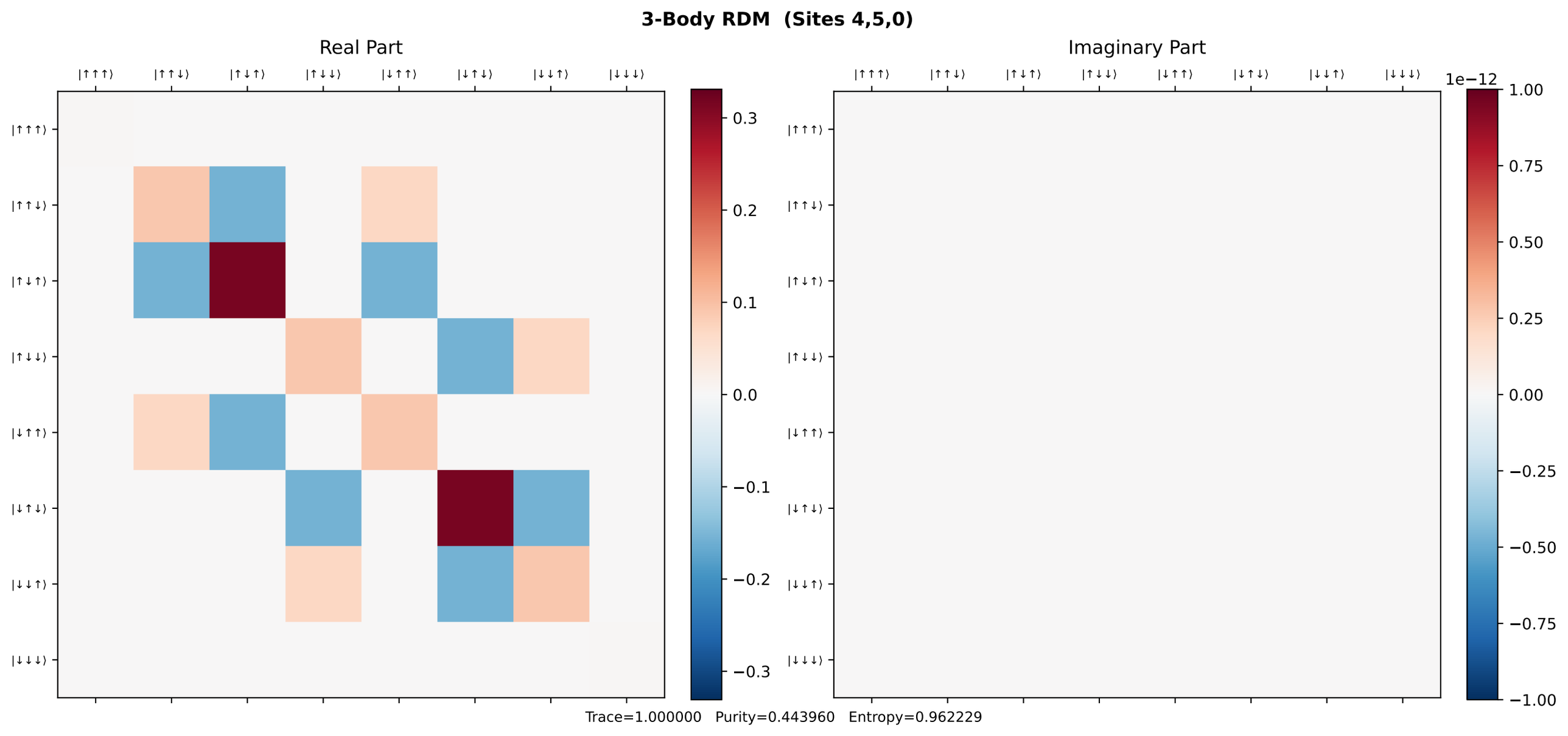
| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 4,5,0)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 4,5,0)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 4,5,0)

## Bipartitions (Sites 4,5,0)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

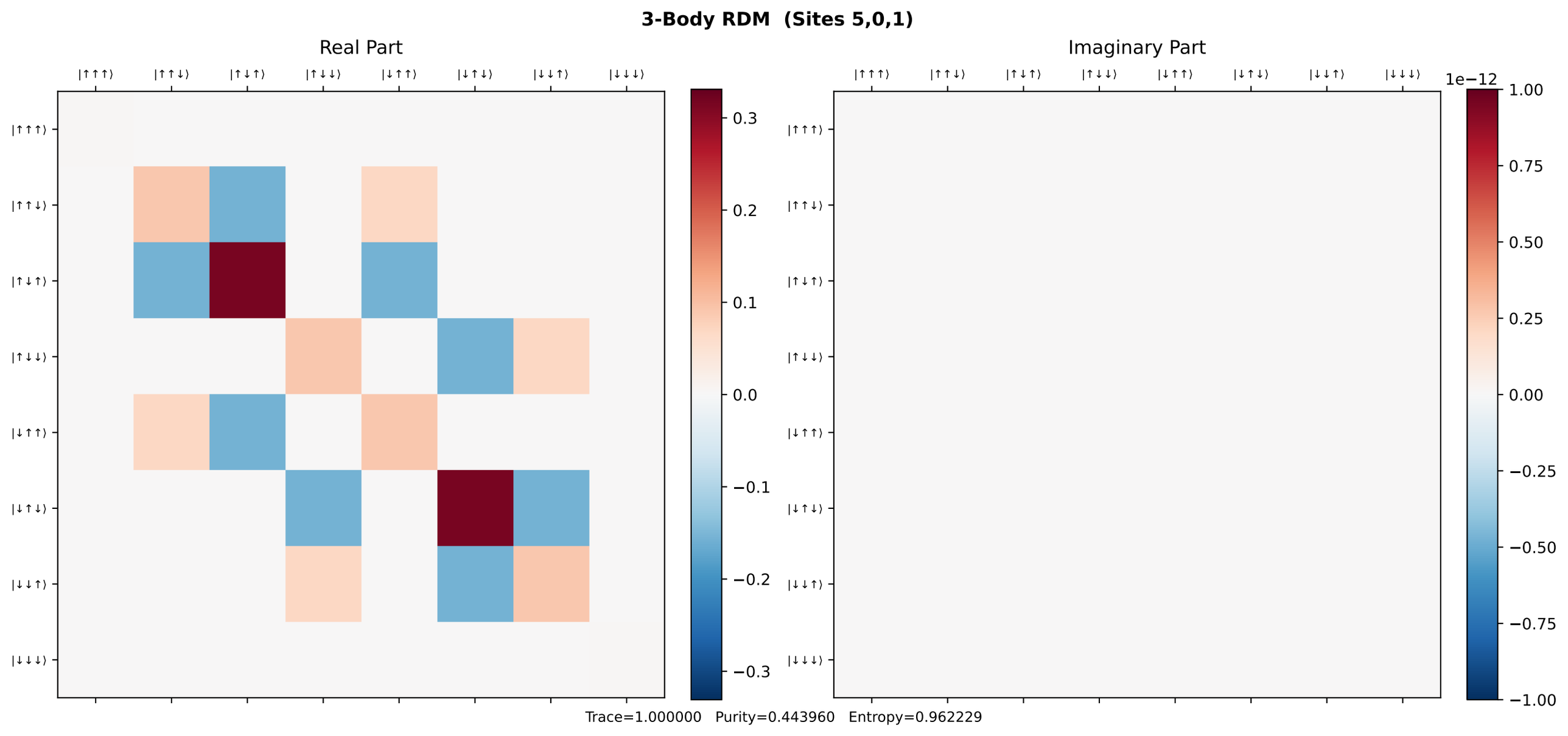
| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


3-Body RDM (Sites 5,0,1)
Real Part
Imaginary Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=1.000000 Purity=0.443960 Entropy=0.962229

```
RDM Metrics  (Sites = 5,0,1)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.4439601755
  Von Neumann entropy    : 0.9622291109
  Eigenvalue  min        : 4.015733e-03
  Eigenvalue  max        : 4.706299e-01
```

# GME Analysis — 3-Body RDM (Sites 5,0,1)

## Bipartitions (Sites 5,0,1)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.44396018$

| A | B | $\mathrm{Tr}(\rho_A^2)$ |
|---|---|---|

## STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).